

DOMANDE DI TEORIA

(2 punti) Un buon documento dei requisiti dovrebbe essere:

A.

Completo e consistente

B. Ambiguo e completo

C. Ridondante e consistente

D. Affidabile e completo

E. Misurabile e completo

(2 punti) Quali sono i benefici del pair programming?

A. Maggiore qualità del codice

B. Maggiore velocità di sviluppo

C. Riduzione dei costi di sviluppo

D. Aumento della conoscenza condivisa

E. Definizione dell'intero design del software nelle fasi iniziali del progetto

(2 punti) Considerando i requisiti non-funzionali di una applicazione software

A. L'affidabilità è più importante dell'usabilità e della manutenibilità

B. L'usabilità è più importante dell'affidabilità e della manutenibilità

C. La manutenibilità è più importante dell'usabilità e dell'affidabilità.

D. La manutenibilità, l'affidabilità e l'usabilità sono ugualmente importanti.

E. La classificazione delle proprietà non funzionali deve essere definita dalle parti interessate.

(2 punti) La funzione A chiama la funzione B, che a sua volta chiama la funzione C. Applicando la tecnica di integrazione bottom-up come si deve procedere?

A. Test di C, poi test di B+C, poi test di A+B+C

B. Test di C, poi test di A+B+C

C. Test di A+B+C

D. Test di C, poi test di B + mock di C, test di A + mock di B, test di A+B+C

E. Test di A + mock di B, test di A+B+ mock di C, test di A+B+C

(2 punti) Quali sono gli elementi chiave di un diagramma di Gantt?

A. Attività

B. Responsabili delle attività

C. Sequenze temporali

D. Budget di progetto

E. Milestone

Quali delle seguenti affermazioni è rappresentativa riguardo agli obiettivi principali del testing?

- ☐ a. Nessuna delle alternative proposte
- ☐ b. Aumentare la complessità del codice
- ☒ c. Rivelare la presenza di difetti nel software
- ☐ d. Dimostrare che il sistema è privo di difetti
- ☒ e. Garantire la sicurezza del sistema

Qual è la motivazione principale per l'uso delle attività di configuration management?

- ☐ a. Migliorare le prestazioni del software
- ☐ b. Nessuna delle alternative proposte
- ☐ c. Ridurre il tempo di sviluppo
- ☐ d. Migliorare la sicurezza del sistema
- ☒ e. Consentire l'accesso concorrente e la modifica dei documenti / codice sorgente

Quali sono le caratteristiche distintive del pattern architetturale "Client-Server"?

- ☐ a. Non esiste un modello di dati condiviso tra i sottosistemi
- ☒ b. I server forniscono servizi specifici come la gestione dei dati
- ☒ c. I client possono accedere ai server tramite una rete

- ☐ d. I dati sono distribuiti in modo uniforme tra tutti i client
- ☐ e. I client chiamano direttamente i metodi dei server senza intermediari

Quali sono le principali attività di produzione nel processo di ingegneria del software?

- ☐ a. Nessuna delle opzioni proposte
- ☒ b. Implementation
- ☒ c. Design
- ☐ d. Configuration management
- ☒ e. Requirement engineering

Quali sono i tipi di proprietà del prodotto software definiti nell'ingegneria dei requisiti?

- ☒ a. Non Funzionali
- ☐ b. Finanziarie
- ☐ c. Organizzative
- ☐ d. Estetiche
- ☒ e. Funzionali

Quali sono i vantaggi dell'utilizzo di un modello di controllo delle modifiche "Lock-Modify-Unlock"?

- ☒ a. Riduzione dei conflitti
- ☒ b. Nessuna modifica concorrente
- ☐ c. Facilità di utilizzo
- ☐ d. Obbliga a serializzare il lavoro
- ☐ e. Maggiore velocità di sviluppo

Quali sono i componenti principali di un documento di requisiti?

- ☐ a. Budget di progetto
- ☒ b. Specifiche di performance
- ☒ c. Prototipi dell'interfaccia utente
- ☐ d. Diagrammi di Gantt

- ☒ e. Descrizione delle funzionalità

Quali sono le pratiche chiave di XP?

- ☒ a. Piccole release
- ☐ b. Design completo a inizio progetto
- ☒ c. Pair programming
- ☐ d. Documentazione esaustiva
- ☒ e. Code ownership appartenente a tutto il team

Qual è uno dei limiti principali dell'ispezione del codice?

- ☐ a. Può essere eseguita solo durante il testing dinamico
- ☒ b. Richiede molto tempo ed è dispendiosa in termini di risorse
- ☐ c. Non è efficace nel trovare difetti funzionali
- ☐ d. Non può rilevare difetti di progettazione
- ☐ e. Non può essere applicata al codice sorgente

Quali attività rientrano nelle attività di validazione e verifica (V&V)?

- ☒ a. Design inspection
- ☒ b. Code inspection
- ☒ c. Requirement inspection
- ☐ d. Configuration management
- ☐ e. Testing

5 Correctness of a software application is defined as

- A. Complete satisfaction of all customers
- B. Complete satisfaction of the majority of users
- C. Actual output corresponds to the expected output for most inputs
- D. Actual output corresponds to the expected output for all inputs
- E. Successful certification by the specific domain authority

11 A 10 person days software project can be completed in at least (one or more answers)

a- 10 days by one person

b- 1 day by 10 persons

c- ½ day by 20 persons

c- 5 days by two persons

d- None of the above

Options b and c are possible in theory, but unreasonable in practice, since it is not possible to parallelize the development among 10 people or more

12 The goal of testing is

a- Showing that a software application works

b- Showing that a software application is correct

c- Finding defects in a software application

d- None of the above

10 In the context of project management a 'milestone' is

a- a measure of completion of a project

b- an event in a project with consequences on the evolution of the project

c- a measure of effort

d- None of the above

9 Regression testing means

a- Testing a previous version of an application

b- Using test suites of a previous version of an application to test a current version

c- Building a test suite using current and previous versions of test cases

c- Rebuilding an application in a previous version and testing it with new test cases

d- None of the above

12 Exhaustive testing means

a- Defining and applying enough test cases until the effort available for testing is exhausted -0.05

b- Defining and applying all possible test cases 1

c- Defining and applying test cases until a very high level of coverage is achieved -0.05

d- Defining and applying test cases until the computing capability of the testing processor is exhausted -0.1

e- None of the above -0.05

11 In an agile process an iteration produces

a- A complete system -0.05

b- A partial version of the final system 1

c- An initial documentation of the final system -0.1

d- None of the above -0.05

10 A project is estimated to require 16.000 person hours. 4 people can work full time on the project, and a working week contains 40 hours. The project will last:

- a- 100 calendar weeks 0.5
- b- 110 weeks 0.05
- c- More than 100 calendar weeks 1
- d- None of the above -0.05

Mathematically $16.000 / (4 * 40) = 100$. However this is a mathematical lower bound, most likely the duration will be ≥ 100 calendar weeks

9 Showing correctness for a software function can be done:

- a-in all cases -0.1
- b-in some cases 1
- c- It is not a meaningful problem, testing can be used instead -0.05
- d- none of the above -0.05

correctness == exhaustive testing, which is feasible in some (rare) cases. See also question 12

4 Focusing on software design, describe the concept of 'coupling' and make an example

See slides

5 Describe briefly the inspection technique applied to source code

See slides

6 The software application (app for user + server side) has been developed custom for the company managing the e-scooter service, over a 12 months period by 3 full time people. The company wants to operate the service for at least 7 years, and plans 1.5 FTE per year for maintenance.

Where do you expect to have the largest part of costs ? (consider 1 year development +7 years maintenance, consider software costs only)

Development $12 * 3 * 1 = 36$ person months = 3 person years

Maintenance: $12 * 1.5 * 7 = 126$ person months = 10.5 person years =

Maintenance costs are (according to these estimates) the major cost source

7 Describe shortly the 'Adapter' Design pattern

See slides

8 The estimated effort for a software project is 1000 person hours. On the project can work 2 people full time, 2 people part time (50%). Make an estimate of calendar duration of the project, and explain it. Assume 8 working hours per day, 5 days per week.

2 people full time + 2 people half time = 3 FTE

One calendar week = $8 \times 5 = 40$ ph

Available ph per week = $3 \times 40 = 120$ ph

$1000 / 40 \times 3 = 8.3$ calendar weeks = approx 2 calendar months

In practice the duration of the project can be estimated to no less than 2 months, likely at least 3 months considering the staffing profile (not all personnel can work in parallel during the whole project)

7 What are the main differences between a waterfall process and an agile process?

Waterfall:

- only sequential activities (no parallelism)
- one big iteration (starting with the requirements and ending with a big integration + delivery)
- big importance to documentation and fixed price business interaction

- suitable for large size projects

Agile:

- parallel activities
- many iterations (of 24 hours and each iteration is within a sprint of 3-4 weeks ending with a delivery)
- not big importance to documentation (time and material business interaction)
- suitable for small projects

See slides

6 Describe the 'pipe and filter' architectural pattern

See slides

5 Considering a software project, what is the relationship between effort and size?

Effort = $f(\text{size})$ where f is linear or more than linear

Productivity as a fixed number implies a linear relationship between effort and size, but this is not always true

4 What is an 'oracle' ? How can it be implemented?

An oracle is a (theoretical) entity capable of providing the correct result of a function, given the inputs. The practical implementation of an oracle is always subject to error. Practical implementations are humans, or other software applications (ex previous versions or legacy version of the same function, or mathematical formulas implemented by other similar programs)

4) (1 point) User Mario commits file A on a subversion server, subversion notifies that the file is at revision 300. User Mario and other users execute 3 more commits of other files. Then John commits again file A. What is the revision number of A?

304 on the repository and on John's working copy. (However, Mario's working copy is at 301, unless Mario updates)

5) (1 points) What are the key points of perspective based reading?

Like an inspection, but each reader reads the document from a different perspective, ex tester, designer, end user

6) (1 points) What is a milestone? Give an example of a milestone in a construction project.

A key event in the project, with possible effect on later activities. Ex, construction permit is granted.

7) (1 point) What are the key points of the Cocomo model?

The Cocomo model links effort (in person months) and size (in delivered source instructions). Three models are provided (for simple, intermediate and complex systems), but in all cases the relation is exponential. The Cocomo model was proposed by B Boehm in 81, and is now replaced by Cocomo2

8) (1 point) What are the key differences between testing and debugging?

Testing tries to find failures, debug tries to discover the correspondent fault(s)

9) (1 point) What states Weinberg's law in the context of testing?

The one who develops an artifact (notably code) is not suitable to test it

4) (1 point) Describe the lock-modify-unlock change management policy, and its advantages/disadvantages.

User puts lock on CI, user modifies CI, user unlocks CI. Until lock is set, only one user can modify the CI. Advantage: no concurrent, incoherent modifications. Disadvantage: bottlenecks on CI needed by many users, deadlocks if user forgets to unlock a CI

5) (1 points) What is an inspection? How does it work?
See slides

6) (1 points) What is a deadline? Give an example of a deadline in a construction project.
Calendar date when a task/deliverable has to be completed/issued. Ex: roof ready by May 31 2009.

7) (1 point) According to the Cocomo model, how are related duration and effort?

$TDEV = 3 PM (0.33 + 0.2 * (B - 1.01))$ with $B=1$ in the simplest model

Writing the formula was not necessary, key point was saying that duration (TDEV) depends on effort (PM) with exponential formula, exponent < 1

8) (1 point) What is a build? What are the related problems?

Process (including compilation, link, possibly testing and other activities) to produce an executable starting from source code modules and libraries. Problems: dependencies among modules, finding and using the right modules in term of type and version. Popular tools: Make, Ant, Maven.

9) (1 point) In the context of verification and validation, describe a static analysis technique
Inspection or dataflow analysis or control flow analysis.

4) (1 point) Function A calls function B, that calls function C. You want to apply top down integration. How do you proceed?

1-Test A using a stub for B

2-Test A + B using a stub for C

3-Test A+B+C

5) (1 points) Class A has been developed and you should test it. What is better, do black box testing only, white box only, or both? Why?

Both. WB and BB have different criteria of writing test cases and, together, higher probability of finding them.

6) (1 points) What states the Weinberg law?

A developer is unsuitable to test his/her code. In general, the creator of an artifact is not suitable to evaluate its quality.

7) (1 point) The cost of fixing a defect in a requirements document is higher if the project is in the requirement phase, or if the project is in the coding phase? Why?

Fixing a defect costs less in the requirements phase, essentially because no other artifacts (design document, code, test cases) exist yet at that stage. So fixing a requirements defect in the requirement phase means only changing one document, instead of many (if fixed in subsequent phases). Think to a house: it costs less to add a window in the project plan (= requirements

analysis, you just correct tables or diagrams) rather than adding it when the house is in construction (= writing code, you must correct code) or it is already built and people is living there (=software is in use, you must write patches).

8) (1 point) What are the most important functions of a configuration management tool?

- Identify and manage parts of software
- Handle whole history of repository
- Handle branches ,configurations and versions of software
- Handle accesses and changes to parts

9) (1 point) A defect can be injected, discovered, removed. Of course not injecting defects at all is the best option. Is this feasible? Argument briefly your answer.

Not injecting defects is not feasible: software is not defect-free for definition, since it's the result of a human activity. Probability of inserting a defect writing or changing code is different from zero (Adams's Law)

4 (1 points) – Give the definition of effort on the context of project management, and its unit of measure

Time spent by staff to perform an activity. Unit: Person * hours

5 (1 points) – Describe the Copy-ModifyMerge strategy for controlling changes

See slides

6 (1 points) – Describe what are mixed revisions in Subversion

Mixed revisions are revisions of a working copy whose files have different revision numbers. Mixed revisions are possible only in working copies and not in the central repository.

Example: Suppose you have a working copy entirely at revision 10. You edit the file foo.html and then perform an svn commit, which creates revision 15 in the repository. Therefore the Subversion marks the file foo.html as being at revision 15. The rest of the working copy remains at revision 10. This is a mixed revision.

7 (1 points) – Describe shortly the perspective based inspection technique.
Inspection where readers use different points of view (ex end user, designer, tester) to read the requirement document.

4 (1 points) – Give the definition of calendar time in the context of project management, and its unit of measure

Days, weeks, months in calendar, either absolute (july 13) or relative (day1, day2 ..)

5 (1 points) – Describe the CCB (Configuration Control Board) approach for change management

See slides

6 (1 points) – what is a Configuration Item?

See slides

7 (bonus question) – In question 2 test cases may depend on the day they are executed. How would you approach this problem from the point of view of regression tests?

In practice every day the test cases should be 'shifted' by one day

For instance on 13 7 2012 we have this test T(4384,4390) → T

On 14 7 2012 we need T(4385,4391) → T

Doing this by hand is error prone and time consuming.

So the tester could write a script to automatically produce inputs to test cases in function of when the test cases were written (ex 13 july) and when they are run (today > 13 july)

T(produceInput(4384, 13 july 2012), produceInput(4390, 13 july 2012)) → T

with

produceInput(4384, 13 july 2012) → 4384 +(today - 13 july 2012)

4 (1 points) – Give the definition of effort in the context of project management, and its unit of measure

Effort is the time taken by the staff to complete a task.

It is measured in person hours (std ieee 1045),

and also person day, person month, person year depend on national and corporation parameters

5 (1 points) – What are the possible types of defects in a requirement document?

Omission/ incompleteness

Incorrect Fact

Inconsistency/contradiction

Ambiguity

Extraneous Information

Overspecification (design)

Redundancy

6 (1 points) – What is a Configuration?

A configuration is a set of Configuration Items (CI), each in a specific version, and it can be seen as a snapshot of the software at certain time. Some CIs may appear in different configurations, and also configuration has a version number

4 (1 points) – Describe briefly the ‘repository’ architectural style

Several applications communicate only through exchange of (standardized) data files. No direct interaction between applications, no call to APIs. Example: Eclipse.

(The question was about software architectures, not about repositories in context of configuration management)

5 (1 points) – Provide an example of two conflicting non-functional properties of a software architecture

Very high precision of computation (ex square root precision 10^{-10}) conflicts with Performance. The higher the precision, the slower the response time.

6 (1 points) – Describe briefly the ‘Scrum’ agile process

Sprints iterations of fixed duration (max 4 weeks) producing a working application in increments.
Ranking of requirements by end user / customer (requirements backlog).
Stand up meeting (15’) every day for coordination.

7 (1 points) – A project is estimated to require 40 person months. Give an estimated range of the required calendar time, and explain how you compute it.

A reasonable team for such a project could be 2 to 5 people. Assuming the team has the same number of people from start to end this gives a calendar range of 20 to 8 calendar months (computed as $\text{person} * \text{months} / \text{persons}$).

A team of 40 people is unreasonable (too much coordination and communication overhead) so 1 month duration would be unfeasible. Similarly for 20, 30 people.

4 (1 points) – Describe briefly the MVC pattern

5 (1 points) – Describe briefly the inspection of a requirements document

6 (1 points) – In the context of configuration management, what is a baseline?

7 (1 points) – In the context of project management, give the definition of 'slack time'

4 (1 points) – Describe briefly the 'layered' architectural style

5 (1 points) – What are the possible type of defects in a requirement document?

6 (1 points) – In the context of configuration management, what is a configuration item?

7 (1 points) – In the context of project management, give the definition of 'milestone'

4 (1 points) – Describe briefly the 'repository' architectural style

5 (1 points) – What are the possible type of defects in a code module?

6 (1 points) – In the context of configuration management, what is a baseline ?

7 (1 points) – In the context of project management, give the definition of 'deliverable'

5 (1 points) – “Cost of the project should be measured in Euros” what kind of requirement is this?

Domain requirement (a kind of NF requirement)

6 (1 points) – Describe shortly the Delphi estimation method.

See slides

7 (1 points) – What is the recommend length of an iteration in an agile project?

4 -5 weeks at most

8 (1 points) – Describe shortly the ‘Strategy’ Design pattern

See slides

4 (1 points) – A project is estimated to require 60 person months for development. What could be the calendar duration for development?

Depends on number of people working, 3 people, 20 calendar months, 4 people 15 months
But there is a limit on the number of people (ex 20 people is unrealistic, so 3 months unfeasible)

5 (1 points) – A software product is developed custom for a bank. Development takes 18 months, operation and maintenance 10 years. Where do you expect to have the larger part of costs?

Maintenance, because 10 years is way longer than 18 months, evolutive, corrective and enhancement maintenance will likely require more effort than initial development.

6 (1 points) – Describe shortly the Function Point method.
See slides

7 (1 points) – Describe shortly the ‘incremental’ software process

See slides about process (this is not about incremental integration)

8 (1 points) – Describe shortly the ‘Adapter’ Design pattern

See slides

4 (1 points) – The size of a project is estimated to be 600 FP. The project is developed in Java, past figures from the company tell that one FP is worth 20 to 30 LOC, and that productivity is 10-15 LOC per person days. How many person days would be required for the project? (show steps followed to compute your answer).

600 FP → 12000 to 18000 LOC

Min effort $12.000 / 15 = 800$ PD, max effort $18.000 / 10 = 1800$ PD

5 (1 points) – A software product is developed custom for a bank. Development takes 18 months, operation and maintenance 10 years. Where do you expect to have the larger part of costs?

maintenance

6 (1 points) – What are pros and cons of the 'lock modify unlock' approach to change control?

7 (1 points) – Describe shortly the 'SCRUM' software process

8 (1 points) – Describe shortly the 'MVC' Design pattern

4 (1 points) – What are Function Points used for? How are they defined?

See slides

5 (1 point) – What kind of problem ‘traceability’ identifies?

Linking requirements to design components / functions to code components / functions to test cases

6 (1 point) – A defect on a sw product signaled by a customer has been allocated to a developer, who has found the corresponding error. What happens next?

Fix the defect, rebuild, test, regression test, decide release to attach the patch to, release

7 (1 point) – Describe the key concepts of the waterfall process

See slides

8 (1 point) – Describe shortly the ‘MVC’ Design pattern

See slides

4 (1 points) – 140 person hours are equivalent to how many person months? Explain your answer

The conversion rate depends on national and company rules. If one person month == 140 person hours then result is of course 1. If one person month == 160 person hours then result is 140/160. And so on.

5 (1 point) – Describe the key concepts of ‘mutation testing’.

A technique to evaluate how good a test suite is. Errors are inserted in the program under test (mutations). The more mutations the test suite finds, the better the test suite.

6 (1 point) – Describe the key steps in an inspection process of a requirement document.

See slides

7 (1 point) – Describe the key concepts of the Scrum process

See slides

5 (1 point) – Describe what is an oracle in testing, and the key problems related to it

See slides

6 (1 point) – List the types of defects that can be found in a requirement document

See slides

7 (1 point) – What are the main differences between an iterative and a waterfall process?

Waterfall only performs one iteration

8 (1 point) – Describe shortly the ‘Singleton’ Design pattern

See slides

4 (2 points) – Describe shortly the Scrum technique

See slides

5 (1 point) – Provide an example of a configuration item

See slides

6 (1 point) — Describe the copy modify merge strategy for controlling changes. List advantages and disadvantages

See slides

7 (1 point) – Describe shortly the Pair Programming technique

See slides

8 (1 point) – Describe shortly the MVC Design pattern

See slides

4 (1 points) – Describe shortly the Test Driven Development technique

Write one test case that fails (using requirements), write corresponding code until test case passes, repeat until all requirements are satisfied and all test cases pass

5 (1 point) –Considering GIT, what are the three project sections that it defines, and how are they used?

Git directory, working directory, staging area

6 (1 point) — In project management, what are the units of measure for duration and effort? And what is the difference between these measures?

Duration: calendar time (hours, days, weeks) – can be relative (two days) or absolute (from 12 7 to 14 7)

Effort: person * hours

As by the definition, duration measures the time needed to complete a project, effort the amount of work needed. Given the effort of a project, calendar time depends on how many people work on it (staffing profile).

7 (1 point) –What measures can be used to evaluate the quality of software?

Number of defects found over a period of time (defect rate, MTBF)

Number of defects / size

User satisfaction (using questionnaires)

(remark the question asked 'measures' not 'properties')

8 (1 point) – Describe shortly the Facade Design pattern

See slides

4 (1 points) – Function A calls function B, that calls function C. You want to apply bottom up integration. How do you proceed?

Test C, test B+C, test A+B+C

5 (1 point) –Considering GIT, what are the three project sections that it defines, and how are they used?

See slides

6 (1 point) — What is the typical lifecycle for a change? (draw states and transitions)

See slides

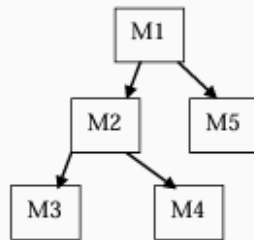
7 (1 point) –What measures can be used to evaluate the quality of software?

Fault density, MTBF, user satisfaction (questionnaire)

8 (1 point) – Describe shortly the Adapter Design pattern

See slides

4 (1 points) – Given this dependency graph propose and justify an integration strategy.



If bottom up: test M3, test M4, test M2+M3+M4, test M5, test all

If top down: test M1 +stubM2 + stubM5, test M1+M2+M5+stubM3 + stubM4, test all

Bottom up requires less stubs / drivers

5 (1 point) – Describe the SCRUM process

See slides

6 (1 point) — In the context of configuration management, explain what is versioning and why it is useful.

Versioning = keep copy of each instance of a configuration item (CI). This allows to keep the history of all modifications to a CI, and allows to roll back to any past instance of a CI

7 (1 point) – Describe a maintenance software process

Receive change requests (CR)

Filter CR

Rank CR

Assign CR to developer – design, code, unit test, integration test

Merge CR with next release of application

release

8 (1 point) – Define the repository architectural pattern. List its pros and cons.

See slides

(question is about architectural pattern, not about configuration management / repository as git, svn etc)

4 (1 point) – In the context of verification and validation, describe Weinberg's law

The creator of a program is unsuitable to test it – for emotional attachment to its creature the programmer tends to overlook defects in it.

5 (1 point) – In the context of change control, describe the lock-modify-unlock technique, its pros and cons.

See slides

6 (1 point) — In the context of configuration management, explain what is a baseline and when is it used.

A baseline is a configuration (== a set of configuration items) in a stable form (ex compiles, links, passes all regression tests).

Not all configurations are baselines

Used to deliver an application internally / externally, while development continues on next version.

7 (1 point) – Describe the waterfall process, its pros and cons.

Activities (requirement design implementation unit test integration test system test) are done in sequence (activity i+1 starts only after activity i is completed).

Document oriented.

Pro: easy structure of activities; agreement on design allows to allocate tasks to many, distributed workers/companies

Con: delivery to customer and validation of requirements and system happen very late. Changes require to restart the process, slowness and lack of flexibility.

8 (1 point) – Describe the singleton design pattern, and when it can be used.

Creational pattern. Ensures that only one instance of the class is created. Ex in an operating system the load balancer must be unique, and could be implemented by a singleton.

4 (1 points) – From the point of view of a user of a software application, what is more relevant, a defect or a failure in the software application? Explain why.

Failure, because user is exposed to it

5 (1 point) – Describe briefly the problem of interactions and trade offs in non-functional requirements

Some NF requirements may be in conflict (ex performance and security), so it is unfeasible to achieve both, and trade offs must be accepted

6 (1 point) — Given a software project with 4 team members, what are the risks if no configuration management is used?

Concurrent access and inconsistent modifications of CIs, unavailability of past versions of CIs

7 (1 point) – Describe the scrum process, its pros and cons.

See slides

8 (1 point) – Describe the abstract factory design pattern, and when it can be used.

See slides

4 (1 points) – In which cases an Oracle can be automatic?

If previous, reliable version of software application is available, or if function can be expressed mathematically

5 (1 point) – Describe briefly ‘mutation testing’ and its purpose

See slides

6 (1 point) — In the context of configuration management, what is the derivation history of a configuration item?

History of versions and changes to them

7 (1 point) – What is the core content of the ISO 12207 standard?

Hierarchical List of activities in the software process

8 (1 point) – Describe the ‘repository’ architectural style, its pros and cons

See slides

4 (1 points) – What are the typical states of a Change Request in a maintenance process?

See slides

5 (1 point) – What is the basic principle of the Visual approach to GUI testing? What is different with respect to layout-based scripted GUI testing?

Visual approach: graphic components (buttons, menus etc) are recognized via image recognition

Layout approach: graphic components are identified retrieving their ID (used by the graphic library) or by specific unique properties.

6 (1 point) -- Provide an example of a static analysis rule from MISRA-C

See slides

7 (1 point) – Describe briefly the pair programming technique from XP

See slides

8 (1 point) – Describe the ‘pipeline’ architectural style, and when it can be used

Many modules are connected in a sequence. Each module is independent of others. Data from one module flow to the next one. Data is the only communication means to a module.

Ex: compiler linker. Ex: unix shell

4 (1 points) – You have to start a software project. What are the factors to consider in selecting a suitable software process?

Criticality (safety critical, mission critical, other), size (correlates to number of developers needed), domain, maintenance vs new development

5 (1 point) – What is the difference between ‘correctness’ and ‘reliability’?

Correctness: a program produces the requested output for every input in the input space

Reliability: probability of providing the requested output over a period of time

Correctness is equal to reliability=100% over an infinite period of time

Normally defects are contained by a program, so reliability is <100% over a finite period of time

6 (1 point) – Describe how versioning is implemented in Subversion.

All Configuration items in a configuration inherit version number from the configuration
Configuration version number increments by 1 (1,2,3,4,...)

7 (1 point) – Describe briefly the SCRUM approach

See slides

8 (1 point) – Describe the ‘repository’ architectural style, and when it can be used

See slides

Remark that the answer is NOT about describing repositories for configuration management (svn, git etc)

4 (1 points) – There are many software processes. What factors characterize a software process?

Number of iterations, sequential vs parallel activities, new development vs maintenance, emphasis on documents

(the question is about *process* characteristics, not *project* characteristics)

5 (1 point) – What techniques can be used to validate the functional requirements of an application to be built?

Inspection of requirements, prototype building, GUI prototype building, writing acceptance test cases

6 (1 point) – In the context of configuration management, what is the purpose of ‘check in’ ‘check out’ operations?

Enforce sequential changes to CI (in lock modify unlock mode) or support parallel changes (in copy modify merge mode) without inconsistencies.

(question is not about what is check in checkout – it is about why they are introduced)

7 (1 point) – Describe briefly the pair programming technique in Xtreme Programming

Two people working on one machine to develop both production code and test cases. One person writes, the other controls and suggests improvements / modifications. The pair reverses roles often.

8 (1 point) – An application is developed for an organization in 6 months by 3 people. Then it is used by the organization during 12 years. Argument whether maintenance costs could be higher /lower than development costs

Maintenance costs will probably be much higher than development cost, since the duration of operations (12 years) is way longer than development (6 months)

Effort for development: $3 \times 6 = 18$ person months

Effort for maintenance: assuming 6 months per year (half a person maintaining the application) = $6 \times 12 = 72$ person months

More precisely, if maintenance requires more than 18 person months / 12 = 1,5 person months per year = 33 person days per year, then maintenance costs will be higher than development costs.

4 (1 points) – Describe the key concepts of Scrum

See slides

5 (1 point) – Given a set of functional requirements, often many designs are possible. How to select one design option versus other ones?

Given that all designs satisfy the functional requirements, selection should be done considering NON functional requirements

6 (1 point) – In the context of configuration management, what is the 'copy modify merge' approach and what are its pros and cons?

See slides

7 (1 point) – What is the definition of 'exhaustive testing' ? Is it possible?

Try all possible test cases (for a function, a class, a program). Normally not feasible due to virtually infinite number of test cases

8 (1 point) – What is 'mutation testing' and what is its goal?

Evaluate how a test suite is good by injecting errors (mutations) in a program and verifying how many mutations are caught by the test suite